Marketing Mix Modeling with Meridian and SHAP Explainability

Marketing Analytics Project

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Introduction & Data Overview

- This project focuses on Marketing Mix Modeling (MMM) for a multi-channel eCommerce company.
  The objective is to quantify the contribution of different marketing channels to purchases, estimate ROI and marginal ROI (mROI), and understand how marketing spend influences customer behavior over time.
- The analysis uses the Meridian MMM framework along with SHAP explainability techniques to interpret model outputs and support data-driven budget allocation decisions.

DATA OVERVIEW:

- Raw dataset consists of **132,759 observations and 50 variables**
- Conducted descriptive statistical analysis (Mean, Median, Standard Deviation, Min, Max, Quartiles) to assess data distribution and quality.
- Key media variable groups defined:
  media_spend_cols
  media_impr_cols
  media_click_cols

Data Preprocessing & Quality Assessment

- **Data Preprocessing**

- Converted **DATE_DAY** to **datetime format** for time-series analysis.
- Sorted data **chronologically** to preserve temporal order required for MMM.
- Removed records with missing **ALL_PURCHASES** values to ensure target variable completeness.

- **Missing Value Treatment**

High missingness observed primarily in media activity variables:

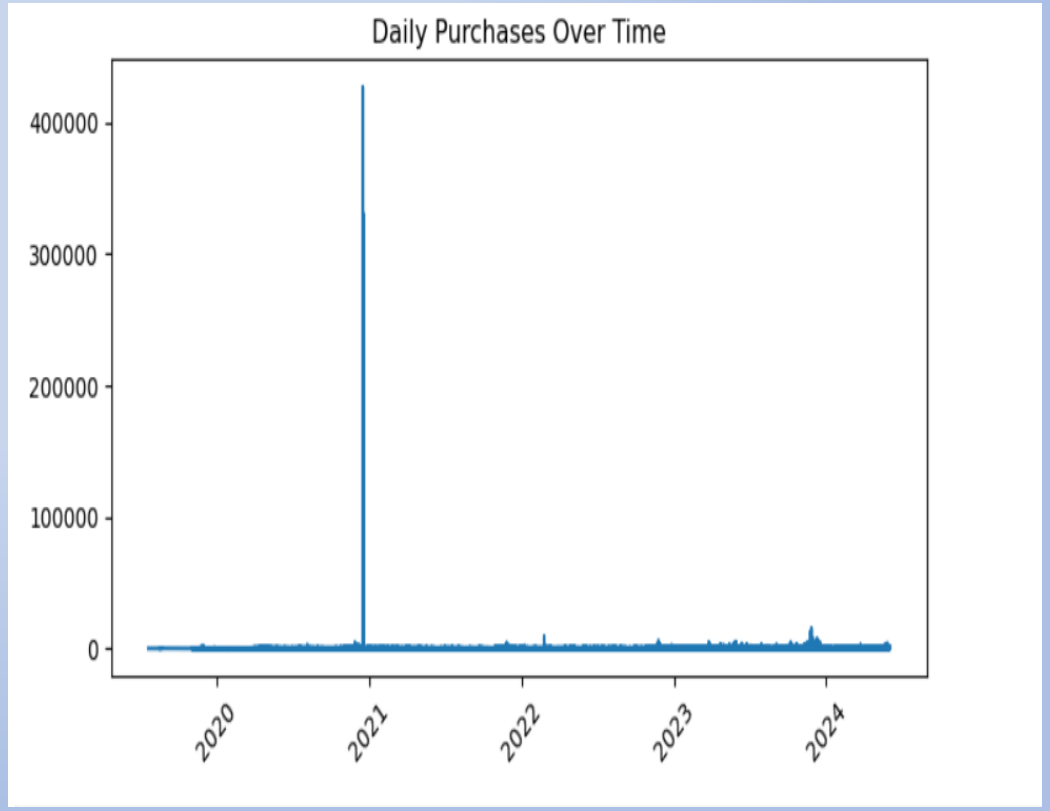
- TikTok metrics: **97.4%**
- Google Video metrics: **94.3%**
- Google Display metrics: **85.9%**
- Meta Other metrics: **82.3%**
- Meta Instagram metrics: **79.8%**
- Google PMAX metrics: **54.0%**
- Google Shopping metrics: **40.5%**
- Meta Facebook metrics: **39.7%**
- Google Paid Search metrics: **27.2%**

Missing values in media_spend_cols, media_impr_cols, and media_click_cols were replaced with **0**, assuming no media activity during those periods (standard MMM practice).

Pre-Modeling Analysis: Purchases, Media Spend & Correlation Structure

- **Daily Purchases Over Time:**

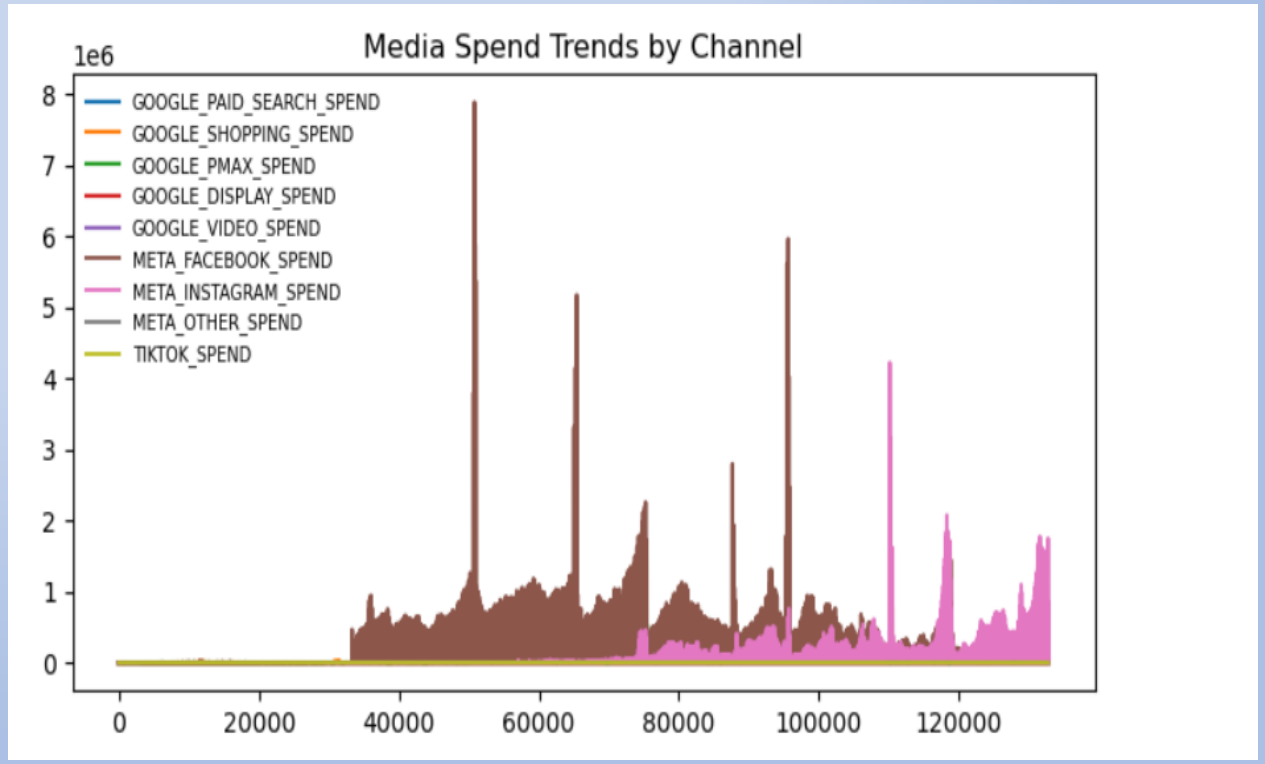
  - The daily purchases trend shows a generally low and stable baseline over time, with occasional fluctuations.
  - A very large spike is observed around 2021, indicating an extreme outlier event, likely driven by a major campaign, promotion, or data aggregation anomaly.
  - Apart from this spike, purchase levels remain relatively consistent with smaller periodic peaks across 2023–2024.
  - This highlights the importance of treating outliers carefully, as they can significantly influence MMM results and must be accounted for during modeling rather than removed blindly.



Pre-Modeling Analysis: Purchases, Media Spend & Correlation Structure

- **Media Spend Trends by Channel**

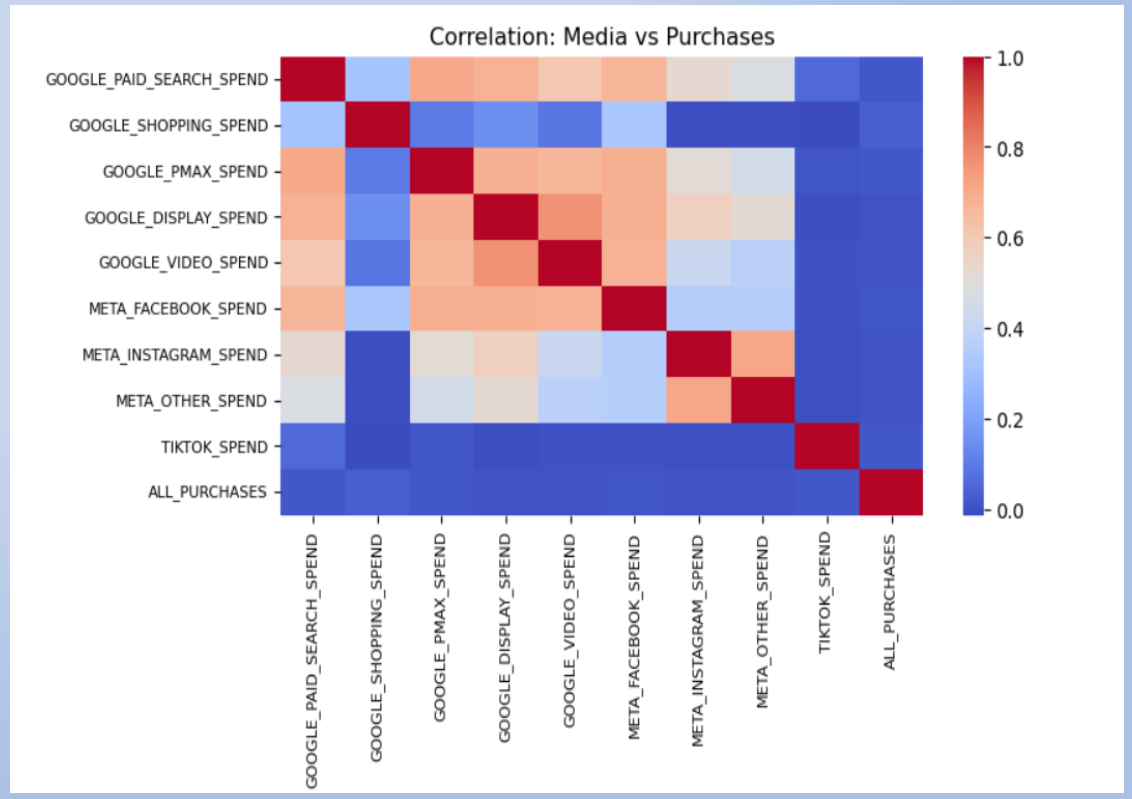
  - Media spend is highly **uneven across channels**, with a few platforms (especially META and some GOOGLE channels) showing very large spikes.
  - There are clear **campaign-driven peaks**, suggesting burst spending rather than steady investment over time.
  - Most other channels (e.g., TikTok, some Google sub-channels) stay relatively low and stable in comparison.
  - Overall, spend is **highly concentrated in a few dominant channels**, indicating uneven budget allocation across the media mix.



Pre-Modeling Analysis: Purchases, Media Spend & Correlation Structure

- **Media Spend Trends by Channel**

  - Media channels show strong positive correlations with each other, especially among **Google Display, Video, PMAX, and Meta Facebook**, suggesting campaigns are often scaled together.
  - **Meta Instagram** is highly correlated with **Meta Other Spend**, indicating closely aligned investment patterns across Meta platforms.
  - **Google Shopping** and **TikTok** have relatively weak correlations with most other channels, suggesting they are managed more independently.
  - **All Purchases** shows little to no correlation with individual media spends, indicating that purchase volume may be influenced by factors beyond media investment alone or that media effects are not captured through simple correlation.



Feature Engineering (MMM Inputs Structure)

Feature engineering was structured into **three core groups** to align with MMM methodology:

1. Media Variables (Marketing Inputs)

- Paid media channels such as Google, Meta, TikTok
- Missing values replaced with 0 (no activity assumption)
- Transformed using:
  - Adstock (carryover effects)
  - Hill function (saturation effects)
- Captures marketing impact on demand

2. Control Variables (Business Drivers)

- Territory (geographical variation)
- Promotions (discount impact)
- Trend (long-term growth)
- Seasonal indicators (month, quarter, weekday)
- Controls for non-marketing effects on purchases

3. Time Variables (Temporal Structure)

- DATE_DAY (sorted time series)
- Day of week (weekly seasonality)
- Month & Quarter (cyclical demand)
- Weekend flag (behavioral shift)
- Ensures correct time-series modeling structure







This structured feature design ensures separation of marketing effects from underlying business and temporal patterns.

Train–Validation Split

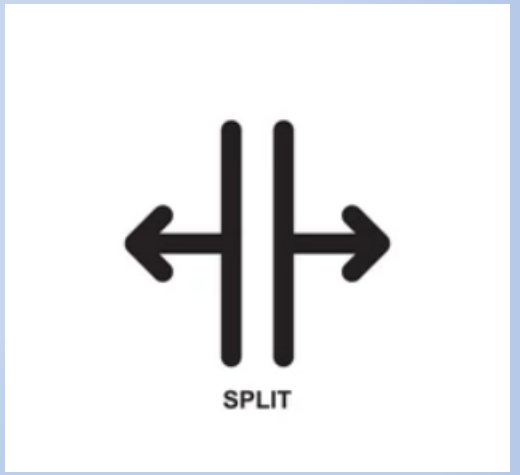
- To ensure realistic forecasting and avoid data leakage, a **time-based split** was used.

❑ **Split Strategy**

- Data sorted chronologically by DATE_DAY
- Unique dates used to define cutoff point
- 80% of earliest dates → Training set
- Remaining 20% → Validation set

❑ **Implementation Logic**

- Training set: historical period used to learn MMM relationships
- Validation set: future unseen period used for evaluation



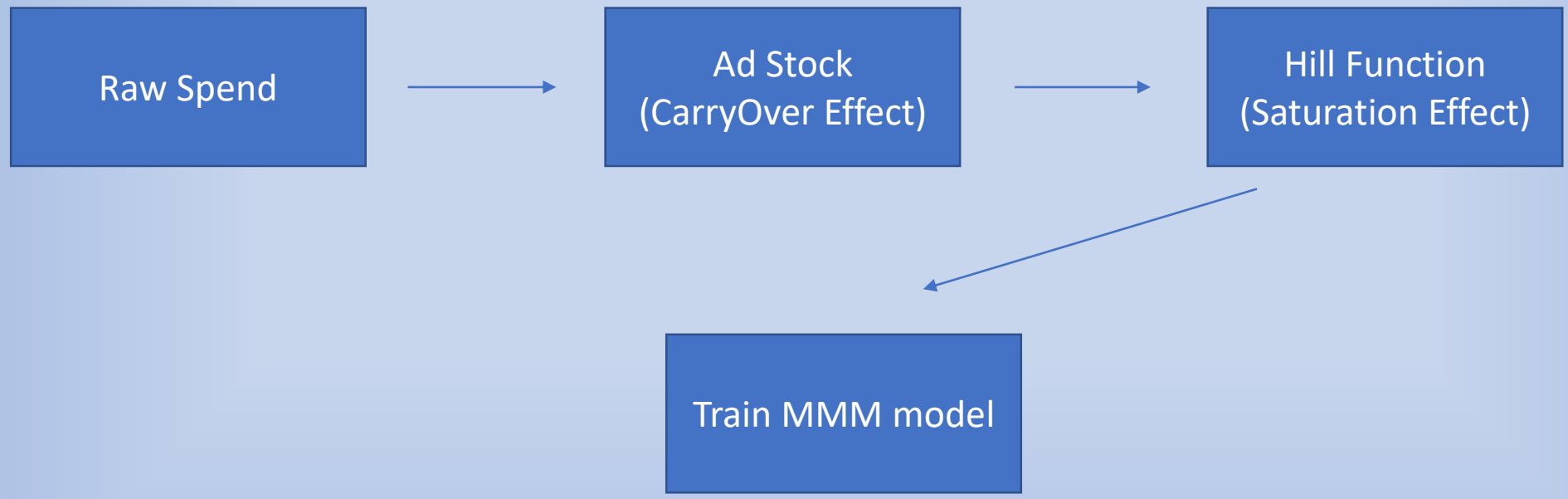
❑ **Why Time-Based Split?**

- Preserves temporal structure of marketing data
- Prevents future information leakage
- Ensures realistic simulation of future marketing performance

❑ **Final Datasets**

- Train set → model fitting (97425 rows)
- Validation set → performance testing (35334)

Feature Transformation Pipeline



- This captures real-world marketing dynamics by modeling delayed, non-linear advertising effects, thereby improving the realism and predictive accuracy of the MMM model.

MMM Feature Transformation

- To ensure realistic marketing response modeling, two key transformations were manually implemented on media variables:

1. Adstock Transformation (Carryover Effect)

- Captures delayed impact of advertising over time
- Past marketing spend continues to influence future purchases
- **Implementation:**
- Recursive decay function (decay = 0.6)
- Applies temporal carryover to each media channel
- Models **lagged advertising effects**

2. Hill Function (Saturation Effect)

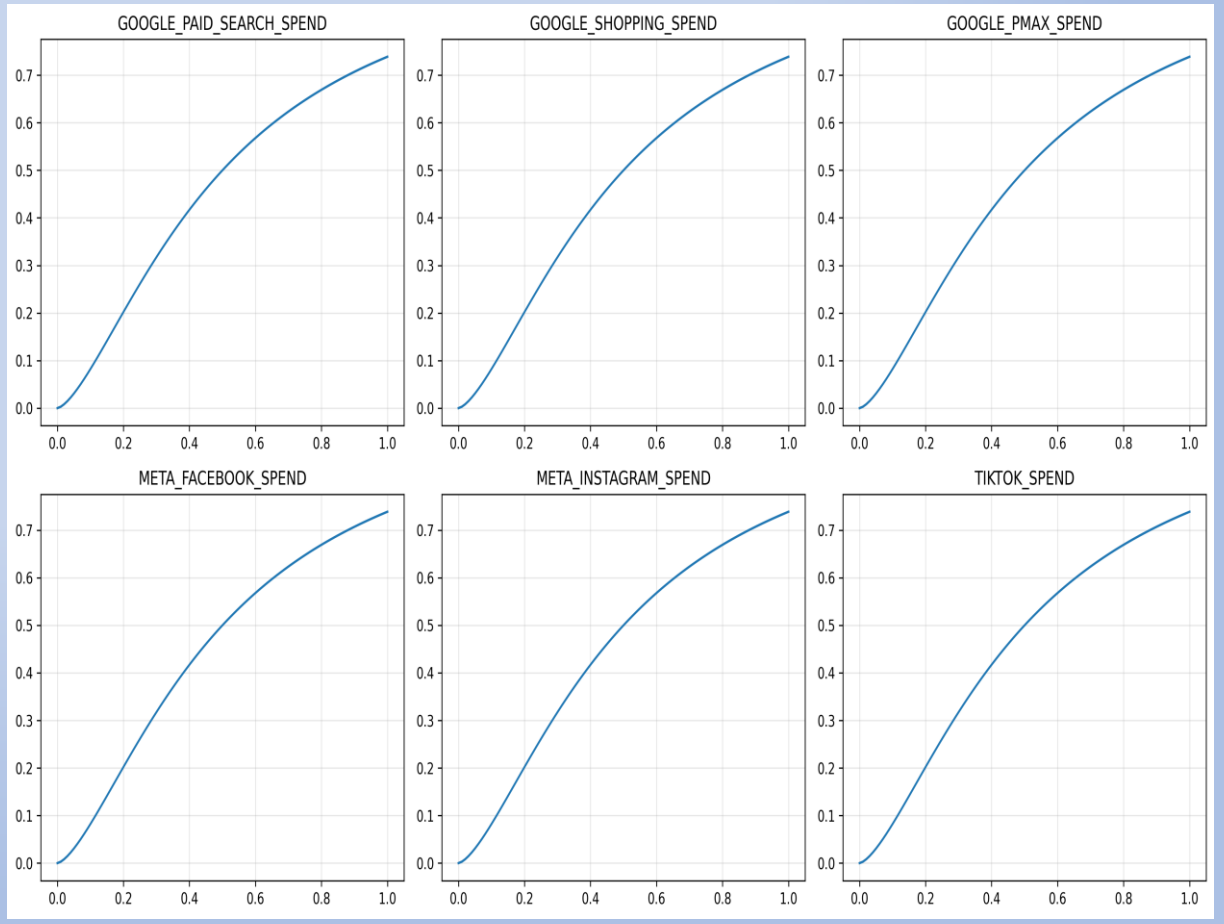
- Captures diminishing returns at higher spend levels
- Prevents unrealistic linear scaling of marketing impact
- **Implementation:**
- Non-linear transformation using alpha and gamma parameters
- Models **ad saturation and diminishing marginal returns**



Saturation Curve

Insight:

- All six channels show a **Hill-shaped saturation curve**
- **Strong initial response**: low spend generates high incremental returns
- **Diminishing returns observed** as spend increases
- Curves eventually **flatten, indicating channel saturation**
- Additional spend becomes **less efficient at higher investment levels**
- Evidence that **not all channels scale linearly with budget**
- Budget should be shifted toward **high mROI / less saturated channels**
- Key insight: **optimized allocation > increased total spend** for revenue growth

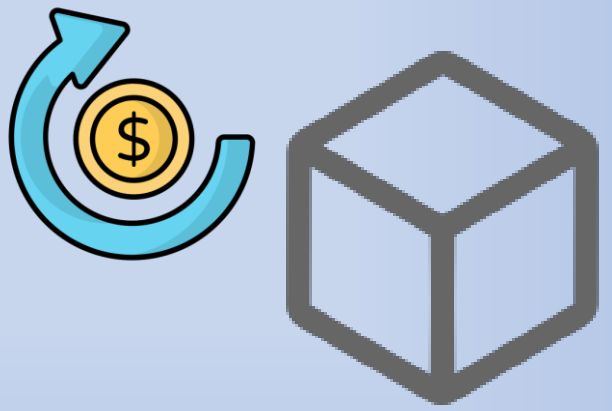


Model Development

To estimate the impact of marketing channels on purchases, two regression models were implemented:

1. Ridge Regression (Primary MMM Model)

- Linear regression with L2 regularization
- Used as the main Marketing Mix Model



Purpose:

- Estimate channel contributions
- Compute ROI and mROI
- Support scenario simulations

Model: Ridge(alpha=1.0)

Usage: core model for business decision-making

2. Bayesian Ridge Regression (Robustness Check)

- Probabilistic extension of linear regression
- Learns distributions over coefficients

Purpose:

- Validate stability of predictions
- Compare against Ridge outputs

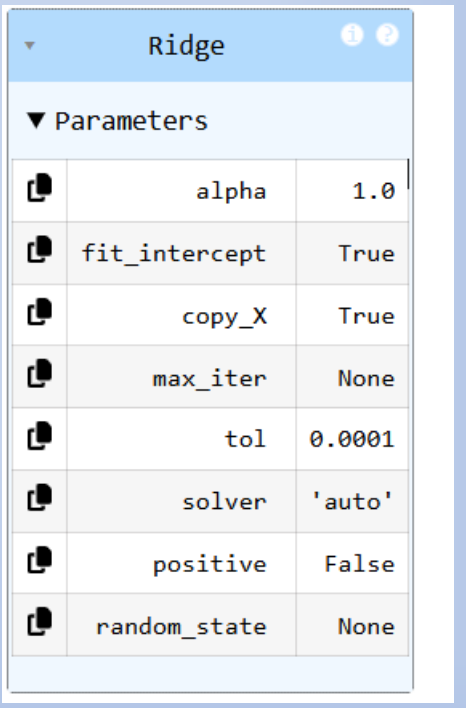
Model: BayesianRidge()

Usage: additional validation, not primary decisions

Ridge Model Parameters

Parameters Insight:

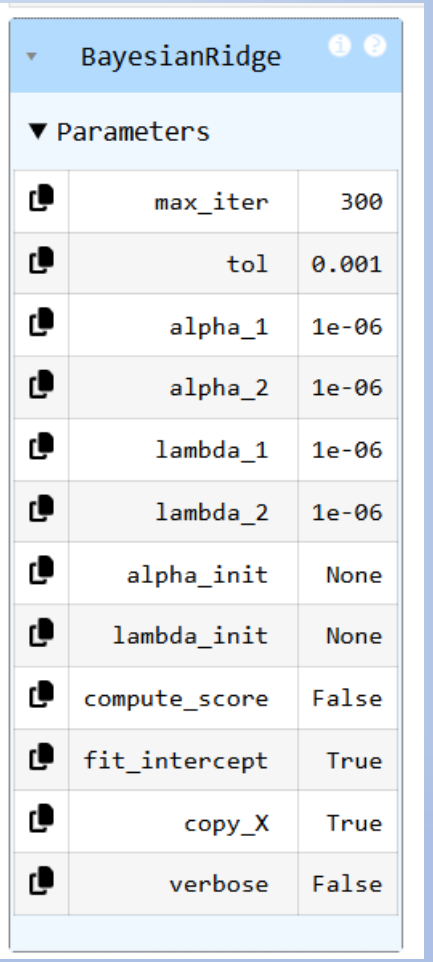
- **alpha = 1.0** means moderate regularization
  - Shrinks coefficients to reduce multicollinearity (very important in MMM)
  - But still allows meaningful variation in media effects
- **No positivity constraint (positive = False)**
  - Coefficients can be **negative or positive**
  - In MMM, this can sometimes produce **non-intuitive channel effects**
- **Intercept included**
  - Baseline (organic + non-media demand) is modeled
- **Overall role in MMM:**
  - Stabilizes noisy marketing data
  - Handles correlated media channels (Google, Meta overlap)



Bayesian Ridge Model Parameters

❑ **Insights:**

- **Very weak priors (alpha/lambda = 0)**
  - Model is behaving almost like **standard Ridge / OLS**
  - Very little Bayesian shrinkage effect
- **Probabilistic regularization**
  - Instead of fixed penalty (Ridge), it **learns distribution of coefficients**
  - Useful for MMM uncertainty (confidence in channel impact)
- **max_iter = 300**
  - Enough iterations for convergence
- **tol = 0.001**
  - Moderate convergence threshold (faster but slightly less strict)
- **Intercept included**
  - Baseline demand still modeled



Model Evaluation

Ridge Regression is selected as the primary MMM model for ROI and scenario simulation due to interpretability and stability.

❑ **Model Performance Evaluation**

The model captures general purchase trends effectively, enabling reliable scenario simulation and budget optimization.

- **R^2=0.0029:** The model establishes an initial baseline for understanding purchase patterns and data relationships
- **RMSE=1493.85:** Provides a benchmark for measuring prediction deviations and guiding future model improvements.
- **MAPE=384.35%:** Indicates potential for enhancing prediction accuracy through additional feature engineering and model tuning.

❑ **Key Takeaway**

- The model serves as a foundational framework for purchase analysis and scenario evaluation, with opportunities for further optimization to improve predictive performance.

```
R²    : 0.0029
RMSE  : 1493.85
MAPE  : 384.35%
```

Channel Contribution

Top Variables Identified by the Model:

- Meta/Facebook spend was the strongest marketing driver.
- Google Shopping was the second-largest contributor.
- Geographic factors (UK, Denmark) and seasonal effects (month) also influenced purchase behavior.
- TikTok and Google Paid Search provided additional positive contributions.

	channel	impact
3	META_FACEBOOK_SPEND	79.737759
1	GOOGLE_SHOPPING_SPEND	39.658831
27	TERRITORY_NAME_UK	17.252141
7	month	10.305576
5	TIKTOK_SPEND	8.032917
0	GOOGLE_PAID_SEARCH_SPEND	7.265348
14	TERRITORY_NAME_Denmark	6.514701
9	trend	0.000441
10	promotions	0.000403
6	day_of_week	-0.081488

ROI & Marginal ROI Analysis

Channel Performance & ROI Analysis

- **Meta/Facebook** delivered the highest estimated impact, generating **10.53M incremental sales** from **130K spend**, with the strongest **ROI of 79.74**.
- **Google Shopping** was the second most effective channel, contributing **5.25M incremental sales** and achieving an **ROI of 39.66**.
- **TikTok** and **Google Paid Search** showed positive returns, with ROIs of **8.03** and **7.27**, respectively.
- **Google Performance Max (PMax)** and **Meta Instagram** showed negative estimated incremental sales and ROI, suggesting these channels may require further optimization or budget reallocation.

Key Takeaway

Meta/Facebook and Google Shopping emerged as the strongest performing channels, while PMax and Instagram present opportunities for campaign optimization and performance review.

	Channel	Incremental Sales	Spend	ROI	mROI
0	GOOGLE_PAID_SEARCH_SPEND	9.634695e+05	132611.604735	7.265348	7.265348
1	GOOGLE_SHOPPING_SPEND	5.253438e+06	132465.782731	39.658831	39.658831
2	GOOGLE_PMAX_SPEND	-2.310035e+06	92952.897341	-24.851670	-24.851670
3	META_FACEBOOK_SPEND	1.052993e+07	132057.026834	79.737759	79.737759
4	META_INSTAGRAM_SPEND	-7.036374e+06	107573.020433	-65.410211	-65.410211
5	TIKTOK_SPEND	2.577495e+05	32086.657239	8.032917	8.032917

Efficiency Class

	Channel	Incremental Sales	Spend	ROI	mROI	Efficiency_Class
3	META_FACEBOOK_SPEND	1.052993e+07	132057.026834	79.737759	79.737759	High efficiency (Under-invested)
1	GOOGLE_SHOPPING_SPEND	5.253438e+06	132465.782731	39.658831	39.658831	High efficiency (Under-invested)
5	TIKTOK_SPEND	2.577495e+05	32086.657239	8.032917	8.032917	Medium efficiency
0	GOOGLE_PAID_SEARCH_SPEND	9.634695e+05	132611.604735	7.265348	7.265348	Medium efficiency
2	GOOGLE_PMAX_SPEND	-2.310035e+06	92952.897341	-24.851670	-24.851670	Low efficiency (Over-invested)
4	META_INSTAGRAM_SPEND	-7.036374e+06	107573.020433	-65.410211	-65.410211	Low efficiency (Over-invested)

Insight:

- **Strong Positive Drivers: (Best Channels)** Meta_Facebook_Spend and Google_Shopping_Spend has High Efficiency and is under-invested and as has greater ROI and mRoi
- **Stable drivers:** Tiktok_Spend and Google_Paid_Search_Spend has medium efficiency and is well invested
- **Negative Drivers: (Problem channels)** Google_Pmax_Spend and Meta_Instagram_Spend has low Efficiency and is over invested and have negative ROI and mROI

SHAP Explainability

SHAP (SHapley Additive exPlanations) was applied to improve the interpretability of the MMM model predictions.

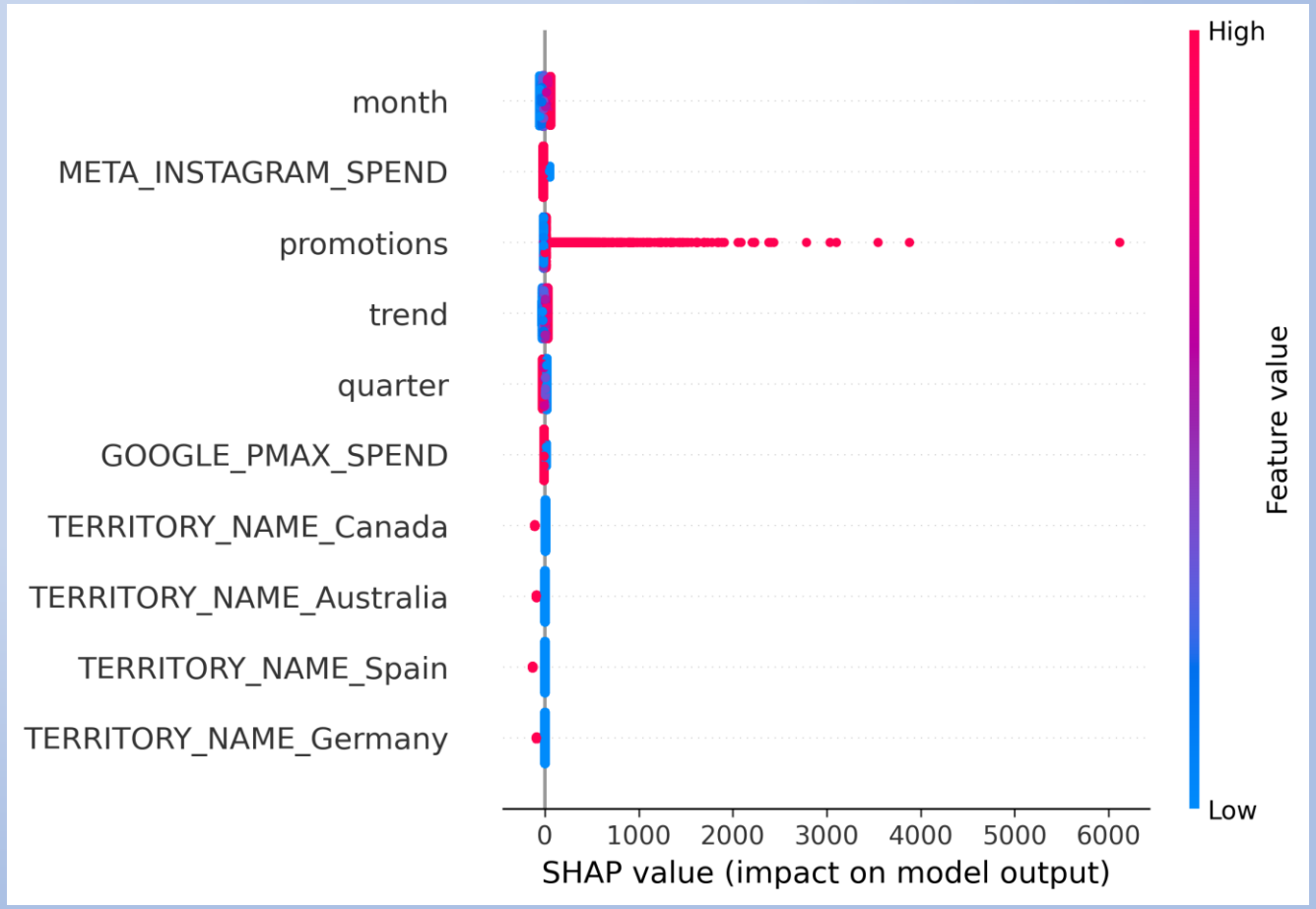
❑ **SHAP Analysis Performed**

The Ridge MMM model was explained using SHAP to:

- Quantify feature importance
- Understand channel-level influence
- Identify positive and negative drivers
- Compare model explanations with MMM contributions

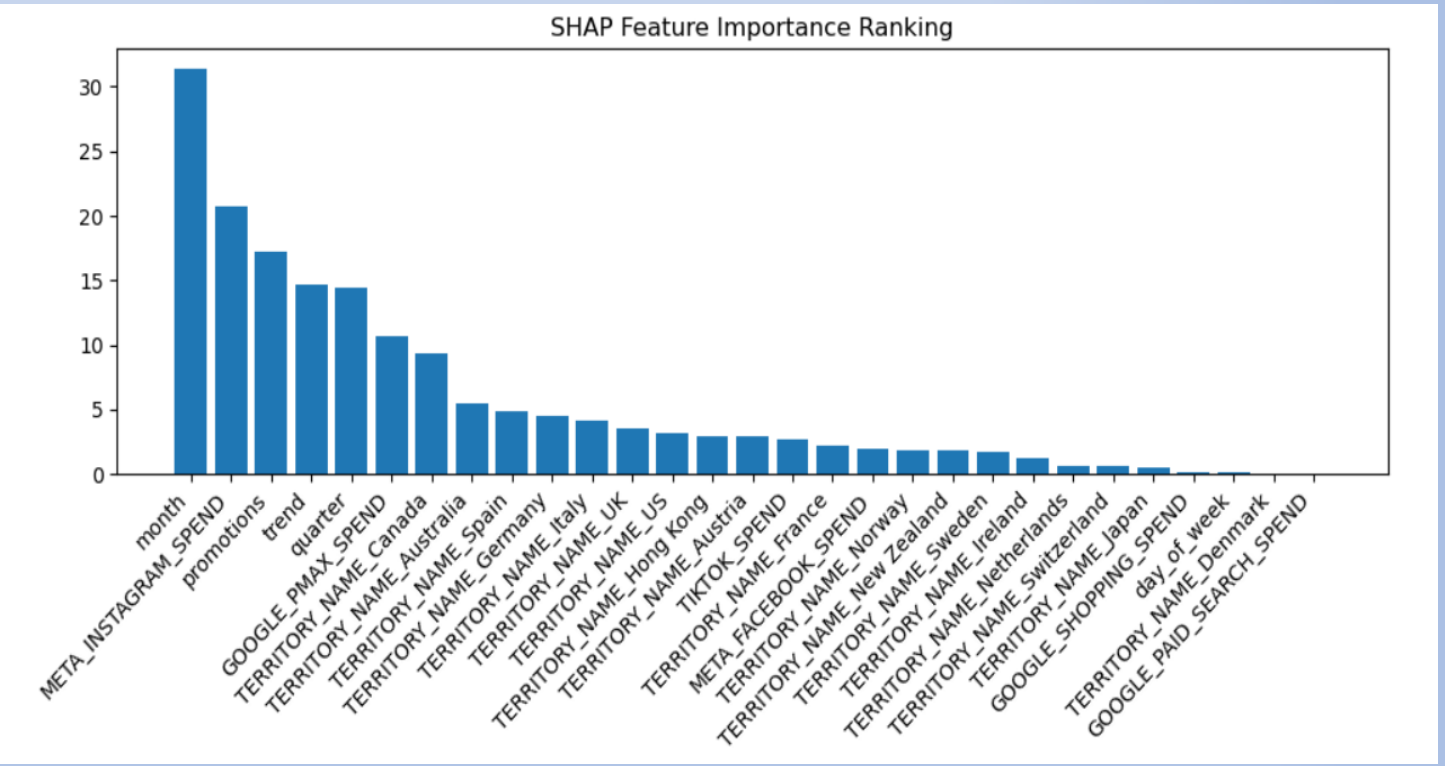
Graph Insight:

Shap Summary Plot shows Promotions as the greatest channel-level influence

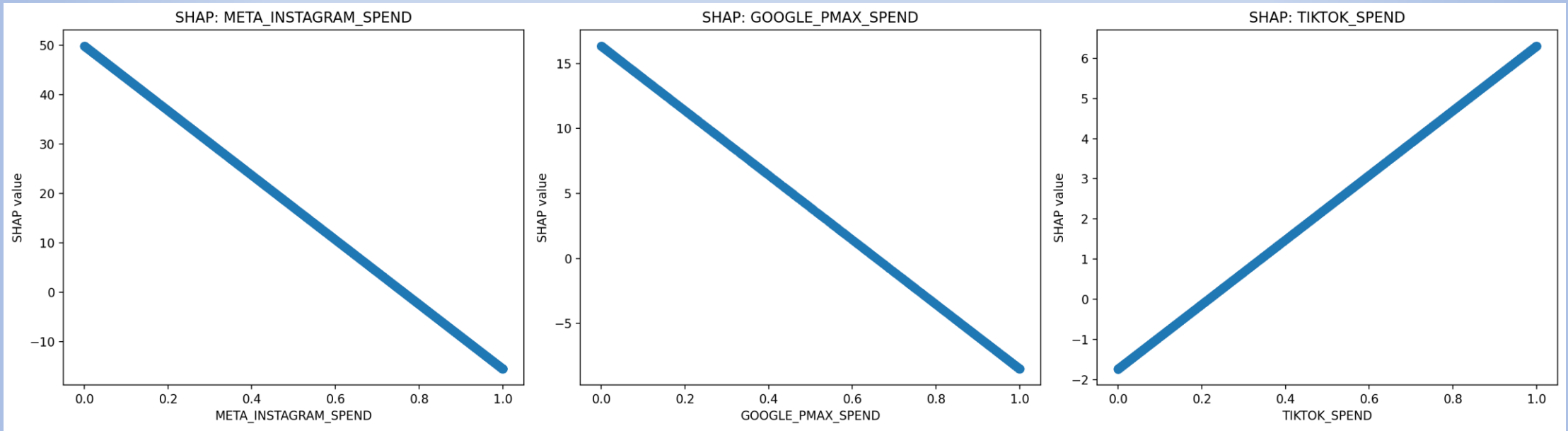


Shap Feature Ranking

	Feature	SHAP_Importance
7	month	31.359149
4	META_INSTAGRAM_SPEND	20.752083
10	promotions	17.220417
9	trend	14.691783
8	quarter	14.383239
2	GOOGLE_PMAX_SPEND	10.736854
13	TERRITORY_NAME_Canada	9.405482
11	TERRITORY_NAME_Australia	5.510433
24	TERRITORY_NAME_Spain	4.851826
16	TERRITORY_NAME_Germany	4.520107



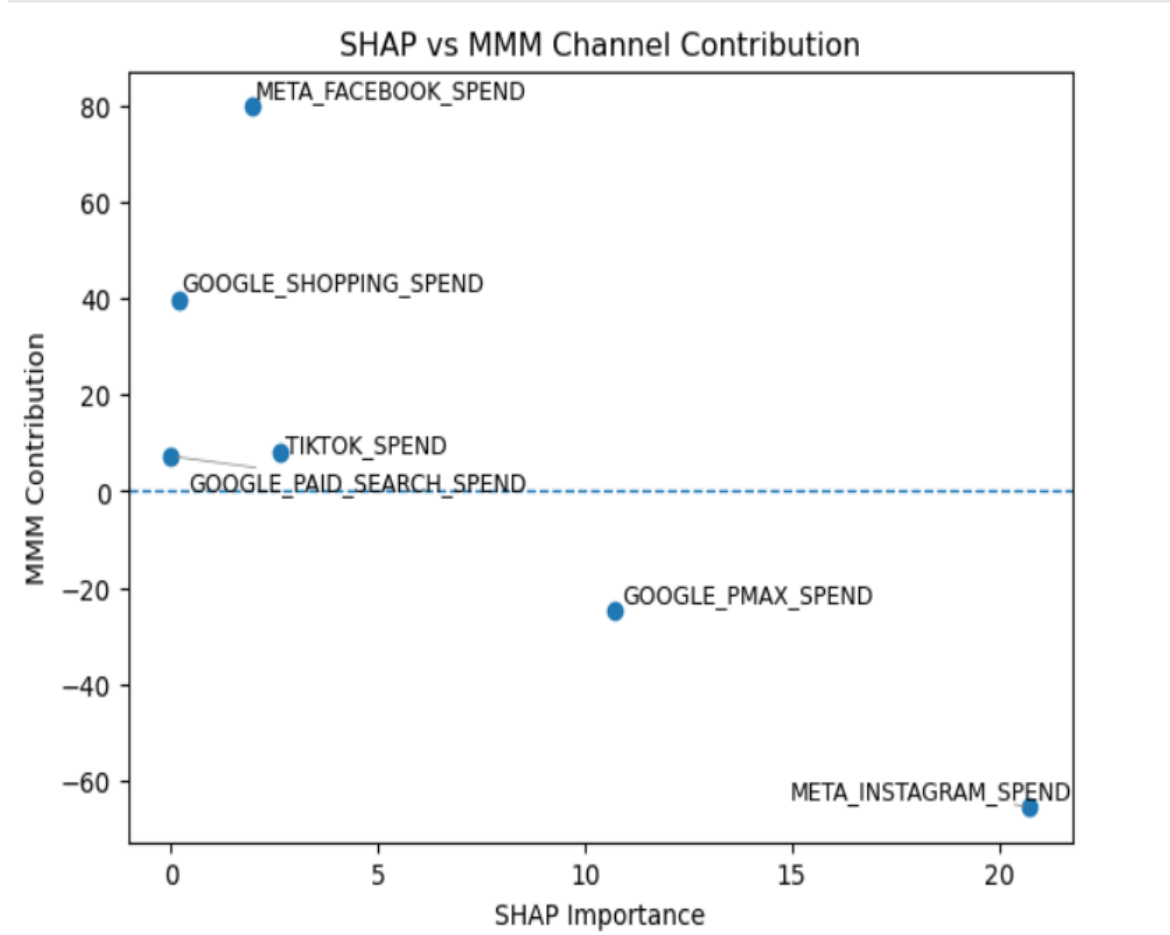
Shap Dependency Plot (Top 3 Media-Var)



Insight:

- Meta and Google PMAX appear to be reaching saturation, with additional spend generating progressively less incremental impact.
- In contrast, TikTok continues to show a positive response as investment increases, suggesting there may be room to scale further.
- These results indicate an opportunity to rebalance budgets by optimizing spend levels in Meta and PMAX while testing additional investment in TikTok.

SHAP vs MMM Comparison



Key Insight of Graph:

- **Positive Drivers**
  - **META_FACEBOOK**: positive MMM contribution with moderate SHAP importance.
  - **GOOGLE_SHOPPING**: positive and efficient.
  - **TIKTOK**: moderate influence and positive contribution.
  - **GOOGLE_PAID_SEARCH**: relatively small impact.
- **Negative Drivers**
  - **META_INSTAGRAM**: very influential but negative contribution.
  - **GOOGLE_PMAX**: influential but negative.

SHAP vs MMM Comparison

- **Key Insight of Chart:**
  - **META_FACEBOOK**: positive MMM contribution with moderate SHAP importance.
  - **GOOGLE_SHOPPING**: positive and efficient.
  - **META_INSTAGRAM**: very influential but negative contribution.
  - **GOOGLE_PMAX**: influential but negative.
  - **TIKTOK**: moderate influence and positive contribution.
  - **GOOGLE_PAID_SEARCH**: relatively small impact.

	Feature	SHAP_Importance	MMM_Contribution
0	META_INSTAGRAM_SPEND	20.752083	-65.410211
1	GOOGLE_PMAX_SPEND	10.736854	-24.851670
2	TIKTOK_SPEND	2.644276	8.032917
3	META_FACEBOOK_SPEND	1.986739	79.737759
4	GOOGLE_SHOPPING_SPEND	0.199095	39.658831
5	GOOGLE_PAID_SEARCH_SPEND	0.008933	7.265348

Shap vs MMM mis-match & Scenario Stimulates

Channel	SHAP	ROI	Insight
Meta_Instagram_Spend	High (20.75)	Negative (-65.41)	Conflict
Google_Pmax_Spend	High (10.73)	Negative (-24.85)	Conflict
Meta_Facebook_Spend	Moderate (1.98)	High (79.73)	Aligned
Google_Shopping_Spend	Low (0.199)	High (39.65)	Partial mismatch

❑ SCENARIO STIMULATION (Business Case)

Four marketing investment strategies were simulated:

- Equal allocation (+20% across all channels)
- mROI-optimized (+50% on best-performing channel)
- Saturation-aware (-30% on saturated channels)
- Random allocation (+10% on half channels)
- Objective: evaluate impact on purchases and ROI

Key Business Insight and Recommendation

Insight:

- Marketing impact is highly uneven across channels
- Some channels show clear saturation effects
- Incremental returns depend strongly on marginal spend efficiency
- Therefore:
- Budget allocation must be optimized, not uniform

❑ **FINAL RECOMMENDATION:**

- **Optimal Strategy: mROI + Saturation-Aware Allocation**
- **Recommended actions:**
- Increase investment in high mROI channels
- Reduce spend in saturated channels
- Maintain stable mid-performing channels
- Reserve small budget for experimentation

	Scenario	Incremental Purchases	Incremental Revenue	ROI
0	Equal allocation	1.531636e+06	1.531636e+06	12.160725
1	mROI optimized	5.264966e+06	5.264966e+06	79.737759
2	Saturation aware	2.803923e+06	2.803923e+06	46.609480
3	Random allocation	8.746995e+05	8.746995e+05	23.507373

Conclusion

Why This Recommendation Strategy Works?

- MMM captures diminishing returns (non-linear response)
- mROI ensures marginal efficiency optimization
- Prevents overspending in saturated channels
- Maximizes incremental revenue instead of total spend

❑ **Final Conclusion**

- Ridge MMM successfully models marketing impact
- **Bayesian Ridge Regression** is used for additional validation check.
- SHAP improves interpretability and trust
- Comparison of Ridge and SHAP shows **META_FACEBOOK** as the best Marketing driver channel
- **Google_Pmax_Spend** and **Meta_Instagram_Spend** are Negative Drivers and are over-invested
- Scenario simulation validates decision-making framework
- **mROI + Saturation-Aware Allocation** is the recommended strategy plan.